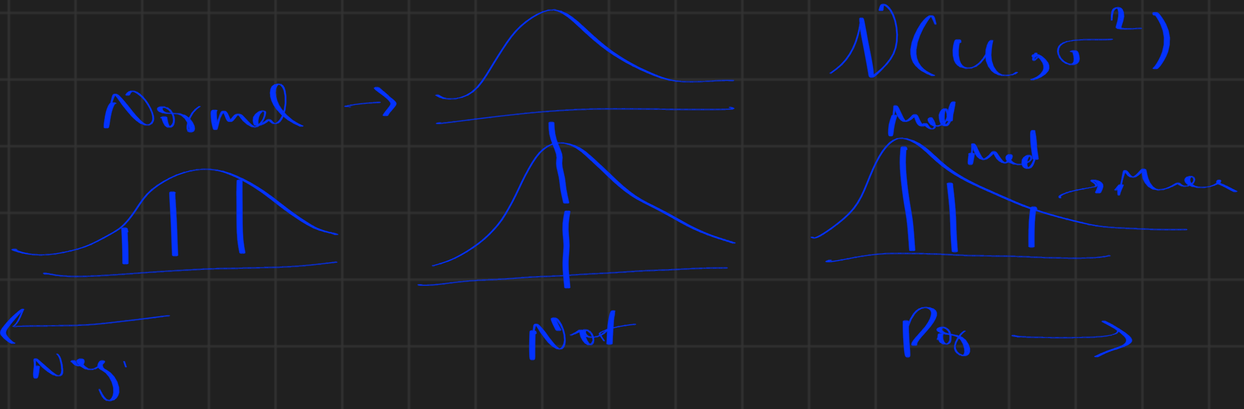
Week 2

→ Descriptive Statistics & Hypothesis testing
→ Guessing Distribution
→ Graphs: P-P plot & Q-Q Plot
→ χ^2 GoF

Mean Median Mode
Var, S.d.
skewness kurtosis

$N(\mu, \sigma^2)$



Coficient of Var (C.V) = $\left(\sigma / \mu\right)$

Std. Nor → $N(0, 1)$ → C.V → $1/0$ → N·

→ Mean = Variance → Poisson(λ)

Hypothesis:

$H_0 \rightarrow \mu_{c_1} = \mu_{c_2}$

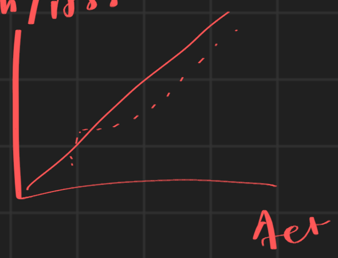
$H_a \rightarrow \mu_{c_1} \neq \mu_{c_2}$

Results : $\begin{cases} H_0 \text{ Rejected} \\ \text{Fail to Reject } H_0 \end{cases}$

Q-Q Plot

→ Quantile - Quantile

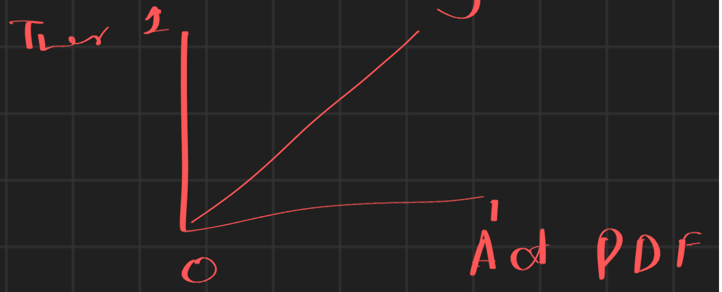
→ Hyp/Ass



→ Sensitive in tails

P-P plot

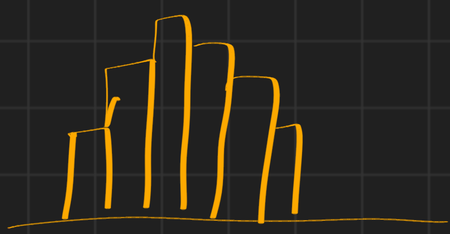
— Probability - Prob.



→ Sensitive in middle region

χ^2 Goodness of Fit

P_1 P_2 P_3 . . . . . P_n



H_0 : Sample follows the Dist.

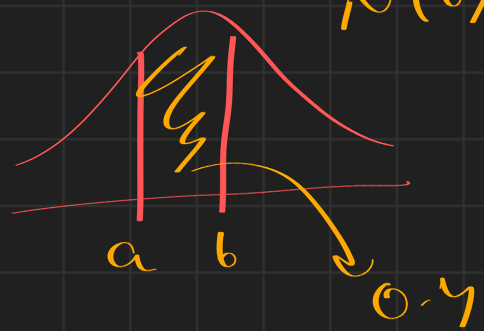
H_a : Sample Doesn't follow the Dist.

Ag: $\boxed{12.5}$, 19, 21, 29.5, 40, 41

B_2 B_2 B_2 B_3 B_3

Bins	Obs fr	Exp. fre
[0-10)	1	$\boxed{0.4} \times 6$
[10-30)	3	0.4×6
[30-60	2	0.1×6
60 -	0	0.1×6

$N(0,1)$



$F_z(a)$

$F_z(b)$

$F_z(b) - F_z(a) =$

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

$\chi^2_{compute}$

$\left\{ \underbrace{\chi^2_{critical}}_{\chi^2_{Tab.}} \right\}$

Tabular < Comp

Reject H_0

Find $\chi^2_{Tab} \rightarrow \alpha$ & dof

dof $\rightarrow N - K - 1$

N ↙ No. of Bins

K ↳ ?

1 ↳ Const

Parameters estimated from sample

$\alpha \rightarrow$ Significance level. $\rightarrow 0.05$

$\Rightarrow 1 -$ Confidence level.

Check if sample follow $N\left\{\begin{matrix}\mu = 32 \\ \sigma^2 = 20\end{matrix}\right\}$

$\hookrightarrow K = 0$

" " " " $N(\mu = 32)$

$\longrightarrow \sigma^2 = ?$

$K = 2$

$\rightarrow$ P & α

$p < \alpha \rightarrow$ Reject H_0

otherwise $\rightarrow$ fail to Reject H_0

P-Value $\rightarrow$ Probability of observing sample when H_0 is true

The P-value is the probability, calculated assuming that the null hypothesis is true, of obtaining a value of the test statistic at least as contradictory to H0 as the value calculated from the available sample data.

Questions.

Q1)

Select the null hypothesis and alternative hypothesis for chi-square test of goodness of fit:

a. Ho: The population follows the proposed distribution
b. Ho: The population does not follow the proposed distribution
c. Ha: The population does not follow the proposed distribution
d. Ha: The population follows the proposed distribution

Q2)

In the Goodness-of-Fit test, if the computed test statistic is greater than the tabulated value of the test statistic at a given significance level, then….

a. Reject the null hypothesis and conclude that the data does not come from population with proposed distribution
b. At the specified significance level, reject the null hypothesis and conclude that the data does not come from population with proposed distribution
c. Do not reject the null hypothesis and conclude that the data does not come from a population with proposed distribution
d. At the specified significance level, do not reject the null hypothesis and conclude that the data does not come from a population with proposed distribution

$\chi^2_{com} > \chi^2_{tab}$

Rej Ho

Q3)

The p-value of the chi-square goodness of fit test represents __

a. The chance of observing the sample when the null hypothesis is false.
b. The chance of observing the sample when the alternative hypothesis is true
c. The chance of observing the sample at the specified level of significance
d. None of the above

Q4)

$p < \boxed{\alpha}$

Reject Ho

If the p-value for the computed test statistic is 0.0524, then at which of the following significance levels, will you reject the null?

a. 1% → 0.01
b. 5% → 0.05
c. 10% → 0.1
d. 15% → 0.15

(Q5)

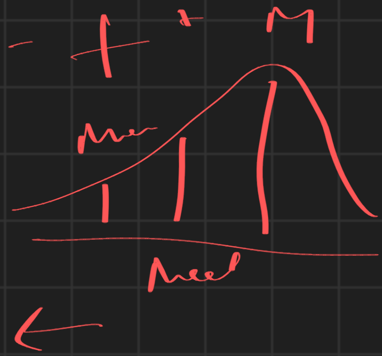
A Distribution is Left Tailed if … MCQ

a. Negative Skewness

b. Left Tail implied mean is on Left Side

c. Mean < Median < Mode

d. All of These



(Q6)

Which among the following is True?

a. QQ plot AMPLIFIES the differences in TAILS of model and sample distributions

b. PP plot AMPLIFIES the differences in TAILS of model and sample distributions

c. QQ plot AMPLIFIES the differences in the MIDDLE PORTION of model and sample distribution.

d. PP plot AMPLIFIES the differences in the MIDDLE PORTION of model and sample distribution.

(Q7)

The table below provides the summary statistic for a random variable. Then what distribution could be a good fit for this random variable? (select all that is/are applicable)

a. ~~Poisson distribution~~
b. ~~Standard normal distribution~~ (0,1)
c. Uniform distribution
d. ~~Exponential distribution~~

Summary Statistic	Value
Number of observations	300
Mean	12
Median	13
Mode	12
Std. Deviation	2
Minimum	11
Maximum	15
Skewness	0.01

→ 12

→ Var - 4

The table below provides the summary statistic for a random variable. Then what distribution could be a good fit for this random variable? (select all that is/are applicable)

a. Poisson distribution
b. ~~Standard normal distribution~~
c. Uniform distribution
d. ~~Exponential distribution~~

Summary Statistic	Value
Number of observations	300
Mean	4.94
Median	5.00
Std. Deviation	2.22
Minimum	1
Maximum	10
Skewness	0.71

$\sigma^2 = \mu$

→ 4.92

Poi = μ = Var = λ

Say, Table 1 specifies the blood sugar level (in mg/dL) for the male and female patients who visit a hospital. It is assumed that the blood sugar level is distributed Normally with a mean of 50mg/ dL and standard deviation of 10 mg/dL. You are asked to perform your analysis by using the following bins. The bins are suggested such that the probability space is split into equal areas.

Bin-1: [0 mg/dL to 50 mg/dL]

Bin-2: [>50 mg/dL to 70 mg/dL]

Bin-3: [>70 mg/dL to 90 mg/dL]

Bin-4: [>90 mg/dL to 100mg/dL]

Φ or F_z

Patient ID	*Gender*	*Blood sugar level (in mg/dL)*	
P1	MALE	71	→ B_3
P2	MALE	84	→ B_3
P3	MALE	76	→ B_3
P4	MALE	78	→ B_3
P5	MALE	89	→ B_3
P6	FEMALE	92	→ B_4
P7	FEMALE	97	→ B_4
P8	FEMALE	95	→ B_4
P9	FEMALE	83	→ B_3
P10	FEMALE	72	→ B_3

	Obs	Exp
B_1	0	2.5
B_2	0	2.5
B_3	7	2.5
B_4	3	2.5
	10 ✓	10 ✓

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

What is the expected frequency in Bin-2 for the statistical test to be performed? *(Note: give only a numerical value rounded to 2 decimal places. For example, if your answer is 10.1253, then input the answer as "10.13")*

→ 2.5

What is the value of the computed test statistic for the statistical test to be performed? *(Note: give only a numerical value rounded to 2 decimal places. For example, if your answer is 10.1253, then input the answer as "10.13")*

→ 13.2

If the p-value for the test is 0.85, then what can be the level of significance if the Null Hypothesis is NOT TO BE REJECTED (select all that is/are applicable)

OPTIONS :

- 0.9
- ✓ 0.15 $\alpha = 0.15 \rightarrow p < \alpha \rightarrow$ × Do No Reject H_0
- Any value between 0 and 1
- Cannot say without more data.

$p < \alpha$

Rje H_0

GoF → DoF: $N - K - 1$

$\rightarrow 4 - 0 - 1 = 3$

0.79 − 0.20

	Obs	Exp
B_1	2	1.4
B_2	1	2.6
B_3	14	11.8
B_4	3	$\boxed{4.2}$
	20	

"Aqua Aqua (AA)" is a manufacturer of water purifiers. AA wants to understand the life of its RO filters. The life of a filter is measured in terms of "Litres of Water Purified (LWP)" before a warning light is raised for maintenance. A sample of 20 water purifiers were tested and the time (LWP) after which the warning light is raised for maintenance of the RO Filter was noted (data provided in Table-2). With this data, AA wants to understand the possible distribution for the life of the RO filter. To this end, it is proposed to bin the data in 4 groups "Low Life (LWP <=25)", "Medium Life (25<LWP<=40)", "High Life (40<LWP<=80)", "Excellent Life (LWP>80)".

The following Φ values are given to you, where Φ(z) indicates the area under the curve (between "0" to "LWP") for a normal distribution with a mean of 60 and standard deviation of 24.

Φ(-1.45) = 0.07; Φ(-0.83) = 0.20; Φ(+0.83) = 0.79; Φ(+1.45) = 0.93

Sample-ID	RO Filter Life (in LWP)	Sample-ID	RO Filter Life (in LWP)
S-1	73 B_3	S-11	70 B_3
S-2	49 B_3	S-12	47 B_3
S-3	12 B_1	S-13	78 B_3
S-4	72 B_3	S-14	15 B_1
S-5	97 B_4	S-15	54 B_3
S-6	95 B_4	S-16	61 B_3
S-7	79 B_3	S-17	91 B_4
S-8	42 B_3	S-18	61 B_3
S-9	70 B_3	S-19	B_2 26
S-10	53 B_3	S-20	64 B_3

Table-2

$\Phi(a) = Ar$

$z = 0.833$

$20/4 \to 5$

If the life of the RO Filter is expected to follow a uniform distribution, then what is the expected frequency in any given bin? *(Note: If your answer is in decimal, enter it rounded to two decimal places. For example, if your answer is "10.256", enter it as "10.26")*

→ 5

If the life of the RO filter is expected to follow a uniform distribution, then what is the value of the computed test statistic? *(Note: If your answer is in decimal, enter it rounded to two decimal places. For example, if your answer is "10.256", enter it as "10.26")*

→ 22

$DoF = N - k - 1 = 4 - 0 - 1 =$

If the life of the RO filter is expected to follow a uniform distribution, then what is the degrees of freedom for the corresponding statistical test to verify the same? *(Note: If your answer is in decimal, enter it rounded to two decimal places. For example, if your answer is "10.256", enter it as "10.26")*

→ 3

If the life of the RO filter is expected to follow a normal distribution with a mean of 60 and standard deviation of 24, then what is the expected frequency in the "Medium Life" bin? *(Note-1: The bins* **DO NOT SPILT** *the sample space into equal areas) (Note-2: If your answer is in decimal, enter it rounded to two decimal places. For example, if your answer is "10.256", enter it as "10.26")*

$P(25 < LWP <= 40) \times 20$

$P\left(\frac{25-60}{24} < z < \frac{40-60}{24}\right) \times 20$

$= P(-1.458 < z <= -0.833) \times 20 \to (0.2 - 0.07) \times 20 = \boxed{2.6}$

If the life of the RO filter is expected to follow a normal distribution with a mean of 60 and standard deviation of 24, then what is the expected frequency in the "Excellent Life" bin? *(Note-1: The bins* **DO NOT SPILT** *the sample space into equal areas) (Note-2: If your answer is in decimal, enter it rounded to two decimal places. For example, if your answer is "10.256", enter it as "10.26")*

4.2

If the life of the RO filter is expected to follow a normal distribution with a mean of 60 and standard deviation of 24, then what is the value of the computed test statistic? *(Note-1: The bins* **DO NOT SPILT** *the sample space into equal areas) (Note-2: If your answer is in decimal, enter it rounded to two decimal places. For example, if your answer is "10.256", enter it as "10.26")*

→ 1.994

$4 - 0 - 1$

If the life of the RO filter is expected to follow a normal distribution with a mean of 60 and standard deviation of 24, then what is the degrees of freedom for the corresponding statistical test to verify the same? *(Note-1: The bins* **DO NOT SPILT** *the sample space into equal areas) (Note-2: If your answer is in decimal, enter it rounded to two decimal places. For example, if your answer is

Assume that a common value of 8.79 is identified for the tabulated test statistic (for both uniform and normal distribution tests), then (choose all that are applicable):

- REJECT the null hypothesis and conclude that the life of the RO filter is not UNIFORMLY distributed
- At the specified significance level, REJECT the null hypothesis and conclude that the life of the RO filter is not UNIFORMLY distributed
- ~~REJECT~~ the null hypothesis and conclude that the life of the RO filter is not NORMALY distributed
- At the specified significance level, ~~REJECT~~ the null hypothesis and conclude that the life of the RO filter is not NORMALY distributed
- DO NOT ~~REJECT~~ the null hypothesis and conclude that the life of the RO filter is UNIFORMLY distributed
- At the specified significance level, DO ~~NOT REJECT~~ the null hypothesis and conclude that the life of the RO filter is UNIFORMLY distributed
- DO NOT REJECT the null hypothesis and conclude that the life of the RO filter is NORMALY distributed
- At the specified significance level, DO NOT REJECT the null hypothesis and conclude that the life of the RO filter is NORMALLY distributed

$\chi^2_{Tab} = 8.79$

Tab < Comp

$\boxed{\text{Reject } H_0}$

	Tab	Comp
Unifor	8.79	22
Normal	8.79	1.99

✓ Ta < Co
Rej H0

Tab ~~<~~ Comp
→ Do Not Rej.